ULTIMATE INTERVIEW HANDBOOK (2026 EDITION)

Enterprise Data Engineering & Agentic AI Mastery

🎯 **TARGET AUDIENCE: MAANG / MANGOS Tier Systems Evaluation**

Core Objective: Master modern enterprise data engineering, Databricks ecosystems, and core Agentic AI platforms in the precise architectural sequence expected by high-bar technical interviewers. This document functions as an advanced reference handbook, fully integrated with the paradigm shifts introduced during the **2026 Data + AI Summit** (including Omnigent open-source agent meta-harnesses, dynamic service runtimes, and Lakebase/LTAP patterns).

MODULE 01 Distributed Data Engineering Foundations

- **Data Engineering Fundamentals:** Core paradigms of high-throughput data pipelines.
- **Distributed Computing Principles:** Network partitions, data synchronization, and processing topologies.
- **OLTP vs OLAP:** Storage layouts, access patterns, and fundamental system trade-offs.
- **Evolution of Architecture:** Data Lake, Data Warehouse, and Lakehouse architectures.
- **Processing Temporalities:** Micro-batching vs stream processing vs pure continuous execution.
- **Data Formats & Compaction:** Deep-dive into Parquet column layouts, Snappy, and Zstd compression profiles.
- **Apache Spark Core Architecture:** Driver and Executor coordination, memory management limits.
- **Execution Plan Mechanics:** Directed Acyclic Graphs (DAGs), Stage boundaries, and Lazy Evaluation rules.
- **Optimizer Deep-Dive:** Catalyst Optimizer phases and Tungsten code generation execution.
- **Adaptive Query Execution (AQE):** Dynamic partition coalescing, join strategy switches, and skew join optimizations.
- **Spark SQL Engine:** Performance tuning, query optimization, and cost-based plan generation.
- **Spark UI Bottleneck Diagnostics:** Spotting and mitigating Data Skew, Disk Spill, Executor OOM, and Shuffle Overheads.

MODULE 02 Enterprise Lakehouse & Lakebase Systems 2026 SUMMIT

- **Delta Lake Internals:** Multi-part transaction logs (`_delta_log`), atomicity protocols, and checkpoint files.
- **Cloud Storage Isolation:** Guaranteeing ACID compliance over object stores via optimistic concurrency control.
- **Time Travel & Audit Trails:** Historical versioning, snapshot isolation, and point-in-time state recovery.
- **Table Maintenance Lifecycle:** Idempotent multi-writer handling, `VACUUM` locks, and `OPTIMIZE` file structures.
- **Data Skipping Stratification:** Traditional `ZORDER` multi-dimensional indexing vs modern Liquid Clustering mechanics.
- **Physical Layer Optimization:** Deletion Vectors for instant updates/deletes and Row Tracking structures.
- **Predictive Optimization Engine:** Fully automated, system-driven catalog file reorganization.
- **UniForm Framework:** Native cross-format metadata mapping layer for zero-copy Iceberg and Hudi interoperability.
- **Databricks Lakebase & LTAP:** Architectural integration of Lake Transactional / Analytical Processing to bypass standard transactional databases.
- **OpenSharing Protocol:** Open-source Linux Foundation standard for zero-copy sharing of secure tables, AI model weights, and encapsulated agent skills.

MODULE 03 Modern Databricks Platform & Contextual Systems 2026 SUMMIT

- **Unity Catalog Infrastructure:** Metastore topology, Catalog boundaries, and multi-cloud shared namespaces.
- **Programmatic Governance Control:** Contextual Row-Level Security (RLS) and dynamic Column-Level Masking.
- **Access Control Paradigms:** Role-Based Access Control (RBAC) vs corporate Attribute-Based Access Control (ABAC).
- **Storage Plane Assets:** Managed vs External Volumes, External Locations, and Cloud IAM Storage Credentials.
- **Compute Virtualization:** Serverless SQL Warehouses (Classic, Pro, Serverless tiers) and the Photon native Vectorized Engine.
- **Lakeflow Architecture:** Serverless pipelines, low-latency native data ingestion queues.
- **Lakehouse & Catalog Federation:** Cross-database zero-copy query virtualization patterns.
- **Genie One Framework & Workspaces:** Real-time text-to-insight compilation using secure runtime code sandboxes.
- **Genie Ontology Engine:** Building a centralized, semantic business graph using the Open Semantic Interchange (OSI) format.
- **FinOps Platform Auditing:** Writing programmatic health audits against core System Tables (`system.billing`, `system.access`, `system.compute`).
- **Databricks Apps & Serverless Micro Apps:** Hosting operational full-stack internal tools natively within the data perimeter.

MODULE 04 Streaming & High-Scale Real-Time Systems

- **Structured Streaming Engine:** Micro-batch vs continuous streaming internals, execution state stores (RocksDB).
- **Time-Variant Data Controls:** Watermarking thresholds, event-time tracking, and handling late-arriving logs.
- **Advanced Stateful Operators:** Custom enterprise state tracking via `mapGroupsWithState` and `flatMapGroupsWithState`.
- **State Management Lifecycle:** Enforcing explicit State TTL (Time-To-Live) to avoid boundless cluster memory exhaustion.
- **Distributed Stream Joins:** Mechanics of Stream-Stream, Stream-Static, and interval-bounded relational joins.
- **Fault Tolerance & Processing Guarantees:** Transactional check-pointing, source replayability, and exactly-once processing pipelines.
- **Custom Egress Topologies:** Designing resilient, transactional multi-destination writers via `foreachBatch` patterns.
- **Auto Loader Engine:** Cloud notification-driven directory parsing and seamless structural schema evolution.
- **Change Data Feed (CDF):** Tracking multi-version mutation logs and log-tailing delta ingestion.
- **Operational Resiliency Patterns:** Orchestrated cluster state recovery, data replay tactics, and heavy historical backfill strategies.

MODULE 05 Enterprise Data Engineering & Lifecycle Ops

- **Medallion Layout Topologies:** Designing clear operational zones (Raw Bronze, Cleansed Silver, Aggregated Gold).
- **Enterprise Mutability Strategies:** Optimization of high-scale `MERGE` workloads and transactional CDC tracking.
- **Temporal Modeling Frameworks:** Implementation of Slowly Changing Dimensions (SCD Types 1, 2, and Type 4 ledger tables).
- **Resilient Ingestion Frameworks:** Designing mathematically idempotent, self-healing pipeline routines.
- **Metadata-Driven Pipelines:** Declarative data engineering using centralized configuration catalogs.
- **Data Contracts:** Enforcing API-style upstream schema schemas, behavioral definitions, and strict structural enforcement.
- **Data Quality & Expectations:** Real-time profiling, automated metric validation rules, and quarantine handlers.
- **Infrastructure-as-Code Integration:** Continuous Delivery via Databricks Asset Bundles (DABs) and Terraform state providers.
- **Orchestration Topologies:** Advanced Databricks Workflows (conditional paths, failure repairs, task parameterization) vs Apache Airflow.
- **Data Observability Matrix:** Implementing OpenLineage hooks, active visual lineage mapping, and proactive SLA/SLO alerting.

MODULE 06 AI-Ready Data Platforms & Feature Infrastructure

- **Enterprise Feature Platforms:** Online/Offline feature stores, unified calculation timelines, and point-in-time join resolution.
- **MLflow Lifecycle Registry:** Advanced model tracking, multi-stage receipts, deployment packaging, and MLflow Recipes.
- **Knowledge Platform Frameworks:** Ingestion platforms optimized for multi-modal unstructured document parsing.
- **Advanced Document Parsing:** High-performance text parsing engines, OCR integrations, and spatial LayoutLM parsing.
- **Structural Chunking Stratification:** Multi-level chunking algorithms (Semantic cohesion, rolling character windows, token-aware bounds).
- **Vector Embeddings at Scale:** Massively distributed embedding pipelines with parallelized inference optimizations.
- **Databricks Vector Search Architecture:** Managed vs External vector indexes, cluster topology, and distance metrics.
- **Incremental Vector Indexing:** Setting up real-time, automated synchronization between mutating Delta Lake states and Vector Search endpoints.
- **Hybrid Retrieval Fusion:** Combining Sparse (BM25 keyword) and Dense (semantic vector) search domains via Reciprocal Rank Fusion (RRF).

MODULE 07 Enterprise Retrieval-Augmented Generation (RAG)

- **Advanced Prompt Engineering:** Structural meta-prompt design, few-shot contextual injection, and programmatic template formatting.
- **Advanced Retrieval Paradigms:** Auto-query generation, parent-child chunk routing, and hierarchical summary expansion.
- **Context Window Compression:** Programmatic summarization filters, information density sorters, and token reduction workflows.
- **Semantic Reranking Architecture:** Utilizing compute-heavy Cross-Encoder models to re-index candidate vectors.
- **Static Verification Guardrails:** Input/Output pattern compliance, safety boundaries, and prompt intent validation wrappers.
- **RAG Evaluation Suites:** Quantitative scoring (Faithfulness, Answer Relevance, Context Recall) via Ragas and TruLens frameworks.
- **Hallucination Defenses:** Automated ground-truth generation and structural self-consistency execution flows.

MODULE 08 Agent Engineering & Autonomous Runtimes 2026 SUMMIT

- **Agentic Topologies:** ReAct execution models, Plan-and-Execute protocols, and dynamic state graphs.
- **Omnigent Open-Source Meta-Harness:** Universal composition abstractions, secure sandboxed execution environments, and runtime agent agility (Claude Code, Codex, Pi).
- **Omnigent Contextual Policies:** Writing state-aware guardrail rulebooks, computing real-time task hazard indexes, and supporting multi-agent collaboration sessions.
- **Agent Bricks Ecosystem:** Enterprise developer workbench integrating frontier foundation LLMs with the underlying Unity Catalog data lake.
- **Tool Invocation & Execution Loops:** Function-calling patterns, state machine reflection, and error self-correction blocks.
- **Model Context Protocol (MCP):** Utilizing MCP standards for uniform tool discovery, safe system prompt access, and data ingestion.
- **State, Memory, and Persistence:** Multi-session state preservation tables, structured conversational state memory, and multi-week long-running agent threads.
- **Human-in-the-Loop (HITL) Routing:** Injecting dynamic asynchronous approval checkpoints for critical system mutations and financial transactions.
- **Agent Execution Trace Observability:** Visual tracking, DAG-style run graphs, and sub-tool latency profiling.

MODULE 09 Enterprise AI Governance & Platform Security 2026 SUMMIT

- **Unified AI Governance:** Model entitlement tracking, lineage-aware AI monitoring, and rigorous system access recording.
- **Gateway-Level Firewalls:** Automated prompt injection prevention, real-time corporate PII scrubbing, and unsafe input blockades.
- **Contextual Service Policies:** Runtime evaluation loops mapping model behaviors against structural data isolation zones.
- **LakeWatch Platform Integration:** Security operations center (SIEM) engineered natively inside the lakehouse to audit end-to-end agent traces.
- **Model Serving Infrastructure:** Provisioned throughput optimizations, auto-scaling execution endpoints, and serverless deployment boundaries.
- **Model Deployment Traffic Controls:** Dynamic real-time token routing, production A/B Testing, Canary releases, and Shadow Model Deployments.
- **FinOps LLM Optimization:** Token caching frameworks, request deduplication, and low-cost distilled model edge routing.

MODULE 10 Systems Design Capstone & Technical Interviews

- **High-Scale Data Systems Architecture:** Designing a 100TB+ sub-minute ingestion and real-time streaming CDC system.
- **Enterprise GenAI Architecture:** Architecting secure, governed RAG with low-latency hybrid index synchronizations at global scale.
- **Disaster Recovery & Redundancy:** Multi-region replication, automated failovers, and transactional log recovery during cloud network failures.
- **Technical System Trade-Offs:** Mathematical cost modeling, memory vs storage optimization, and precision compute provisioning.
- **Performance Tuning Paradigms:** Advanced optimization scenarios across massive clusters under intensive distributed workloads.
- **Whiteboarding Simulation Layouts:** Navigating high-pressure structural system design loops tailored for MAANG-tier organizations.

HANDBOOK CORE OUTCOME & FOCUS

By systematically reviewing this curriculum blueprint, the learner establishes deep technical proficiency across distributed systems, transaction log engines, contextual data governance, and agentic orchestration platforms. This structure is precisely tailored to pass the rigorous Architecture, Platform Design, and System Scale interview loops standard at elite technical organizations like Apple, Meta, Google, Amazon, Netflix, and top-tier AI platform giants.